

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457827.7474 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.057
Transit depth (Rp/Rs)^2	3.02 +/- 1.99 [%]
Orbital Inclination (inc)	79.81 +/- 2.81
Airmass coefficient 1 (a1)	1.18 +/- 0.17
Airmass coefficient 2 (a2)	-0.12 +/- 0.11
Scatter in the residuals of the lightcurve fit is	14.29 %
Best Comparison Star	#2 - [154, 368]
Optimal Aperture	5.39
Optimal Annulus	13.49
Transit Duration (day)	0.039 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.057 vs 0.1594 (deviation 0.26σ)
Duration fit vs published	0.039 d vs 0.0512 d (deviation 0.72σ)
Mid-time O–C	1.4 min
Depth SNR	3.1
Residual scatter	14.29 %

Light curve

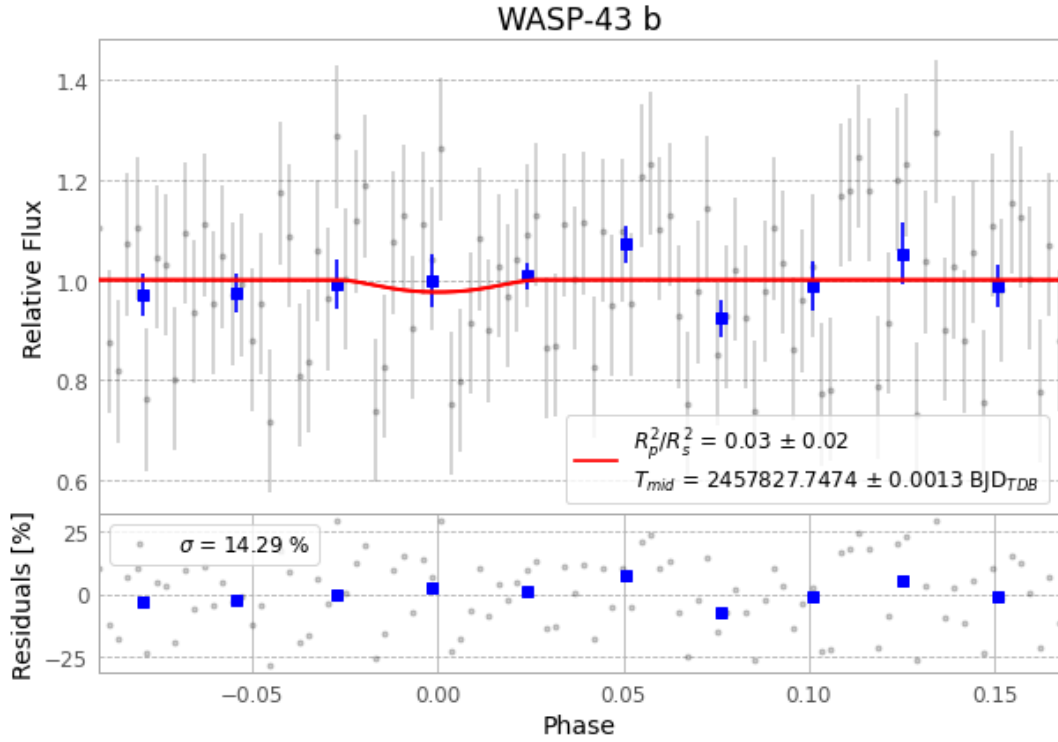


Figure 1 — FinalLightCurve_WASP-43 b_2017-03-14.png (SHA-256 61229abe6e69b642...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [463, 150], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 842cb369599b7bf9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

102 FITS frames (67 MB), WASP-43170315040012.FITS → WASP-43170315090312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-170315.log (523bfd25f4ea...), FinalLightCurve_WASP-43 b_2017-03-14.png (61229abe6e69...), FinalLightCurve_WASP-43 b_2017-03-14.csv (695740e8cea2...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 20:43Z.